

Efficient Thinking: Resource Allocation in Game-Playing AI

Spending Parameters, Data, Search, and Latency Efficiently for Maximum Playing Strength — an Empirical Framework, Demonstrated in Chess

Louay Alsakka · July 8, 2026

A white paper on parameter-efficient game AI.

Code & trained model: github.com/louayalsakka/efficient-thinking

The recurring theme is a set of tradeoffs: memory vs compute (Stage 1 vs 2), speed vs score (the search cascade), and diversity vs correlation (the committee) — each one a different way of spending a fixed budget to buy strength, and each capped by the same wall: the quality of the learned evaluator.

Abstract

This paper asks one engineering question: given finite resources — parameters (memory), training data, inference-time search, and latency — how do you spend them *efficiently* to reach a target playing strength? We treat chess as a clean, fully-measurable testbed and map the tradeoff curves directly, resource by resource. With a deliberately tiny model on modest hardware, a **14 MB** convolutional evaluator (3.45M params) plays at ~2150 Elo as a single forward pass; adding **Monte-Carlo Tree Search (MCTS)** lifts the *same weights* to ~**2800** with zero extra parameters — strength bought with **compute, not size**. That single result is the thesis in miniature: the same capability can be purchased with very different resource mixes, and the efficient mix is rarely "more parameters". (*If you remember three numbers, remember these: a 14 MB evaluator, ~2800 with search, and 4.8× search efficiency from the cascade.*)

The primary contribution is an empirical *framework* for allocating finite AI resources — parameters, data, search, and latency — to a fixed capability goal, with measured tradeoff curves for each; the staged MCTS cascade is its sharpest concrete demonstration. Three specific results support it. **(1) A wide→narrow MCTS cascade** that funnels the simulation budget through progressively narrower, deeper stages

matches flat MCTS at up to $\sim 4.8\times$ less compute per move — our most practical result. **(2)** A direct **MCTS-vs-fixed-depth** comparison: adaptive search both beats alpha-beta at equal compute and *keeps scaling*, while fixed depth plateaus. **(3)** A **teacher-free self-learning** study (self-play, a self-referential ladder, evolution, committees): one robust positive — **agreement predicts correctness**, a confidence signal needing no oracle — and honest negatives — none of these crosses the self-play plateau, and plurality voting does not de-bias. A **capacity sweep** agrees: adding parameters at fixed data ($1.4\times$, $3.45\text{M}\rightarrow 4.81\text{M}$) moves strength ~ 0 , while $8\times$ more *data* moves it $\sim +90$ — capacity is the *weakest* lever in this regime (a full $1\times/1.4\times/2\times/4\times$ sweep on the complete dataset is in progress to confirm it). (A small-scale statement: AlphaZero-scale self-play bootstraps far past its start; we characterize the regime we could run.)

The unifying principle — the **evaluator–search decomposition** — is **strength = evaluator $\times$ search**: search sets how *closely* you approach the evaluator's ceiling; the evaluator sets *where* that ceiling is. And across all three stages the binding constraint consistently emerged as **the quality of the information reaching the evaluator** — its training signal — rather than the machinery around it. **Within a move, search converts the evaluator's latent information into stronger decisions through computation (cutting variance, not bias); it cannot create information the evaluator never learned.** Voting and merging fail for the same reason — they only reorganize what the net already encodes. **Self-play and evolution are different in kind:** they *could* inject new information (search reveals the environment's ground truth via lookahead), but **at our compute they did not exceed supervision — a scale limit, not a theoretical one.** AlphaZero and Leela decisively break past human play with orders of magnitude more self-play than two Mac Studios can run; our plateau characterizes the regime we could reach, and is **not** a claim that self-play fails. The method is as transferable as the numbers: **at each stage a single lever binds — only an experiment reveals which — so effort on any non-binding lever returns almost nothing.**

Headline: *A 14 MB evaluator plus adaptive search reaches ~ 2800 -class play, and a staged MCTS cascade recovers up to $4.8\times$ compute at little Elo cost — thinking, not growing. Search converts what the evaluator already knows into stronger play; it cannot lift the ceiling set by the quality of the information the evaluator was given.***

Read the numbers carefully. Absolute Elo is measured against a Stockfish ladder and carries $\pm \sim 100$ **systematic uncertainty** near the top rung — " ~ 2800 " is an efficiency indicator, not an engine-matching claim, and it is the single easiest figure for a skeptic to attack. The **robust** results are the relative, same-ladder ones: MCTS beats and out-scales fixed depth, the cascade holds Elo while cutting compute up to $4.8\times$, and every Stage-3 aggregation/self-learning method fails to beat a single model. Treat those as the paper's claims; treat 2800 as a headline.

Established results vs. regime-limited observations. We separate the two explicitly:

- **Established** (robust, relative, same-ladder): the **cascade's compute efficiency**, **adaptive search out-scaling fixed-depth**, and **agreement predicts correctness**. These are same-ladder comparisons that do not lean on absolute Elo.
 - **Observations limited to our compute/scale**: the **self-play plateau**, the **flat parameter scaling**, and **evolution's non-escape**. These are true *in our regime* and we expect them to change at AlphaZero/LLM scale — they are honest negatives, not universal laws.
-

Key Takeaways

Five findings — enough to understand the paper on one page.

1. **Search dominates parameters in this regime.** On a *fixed* evaluator, adding search buys **+286 Elo** over the raw policy and keeps climbing **~+55 per doubling** of simulations; *doubling the parameters at fixed data added ~0*. Strength came from **thinking, not growing**.
 2. **A wide→narrow MCTS cascade recovers up to 4.8× compute.** Funnelling the simulation budget through progressively narrower, deeper stages **matches flat-MCTS strength at a fraction of the per-move cost** — our most practical result.
 3. **Adaptive MCTS out-scales fixed-depth search.** At equal compute MCTS **beats** alpha-beta and **keeps scaling**, where fixed-depth search plateaus.
 4. **Model agreement predicts correctness.** Where independent models agree they are more often right — a **teacher-free confidence signal**, and a result independent of chess.
 5. **Find the binding bottleneck experimentally.** At each stage exactly one lever binds — capacity, search, data, or the quality of self-generated signal — and effort on any *other* returns almost nothing. The transferable contribution is this **diagnostic**, not any single Elo number.
-

1. Introduction

Every AI system faces an economic problem. Capability is purchased with finite resources — model capacity (memory), training data, inference-time computation, latency, and human supervision. The engineering objective is not to maximize any one resource but to **maximize capability per unit cost**: more capacity, data, search, or supervision each raise strength, but each carries a cost and to a large degree they *substitute* for one another. This paper presents an **empirical framework for measuring those tradeoffs**, using chess as

a clean, fully-observable testbed — we fix the objective (playing strength) and measure what each resource buys, and where spending on one is wasted because a *different* resource is the binding constraint. We claim **no solved optimal-allocation rule**; the contribution is the framework and a set of *measured tradeoff curves*.

The argument is a three-level hierarchy: **(1) AI resource allocation** — the main thesis; **(2) the evaluator–search decomposition** — the analytical model of how capacity and search combine into strength; and **(3) binding-bottleneck analysis** — the diagnostic that identifies, at any stage, which single resource to spend on next.

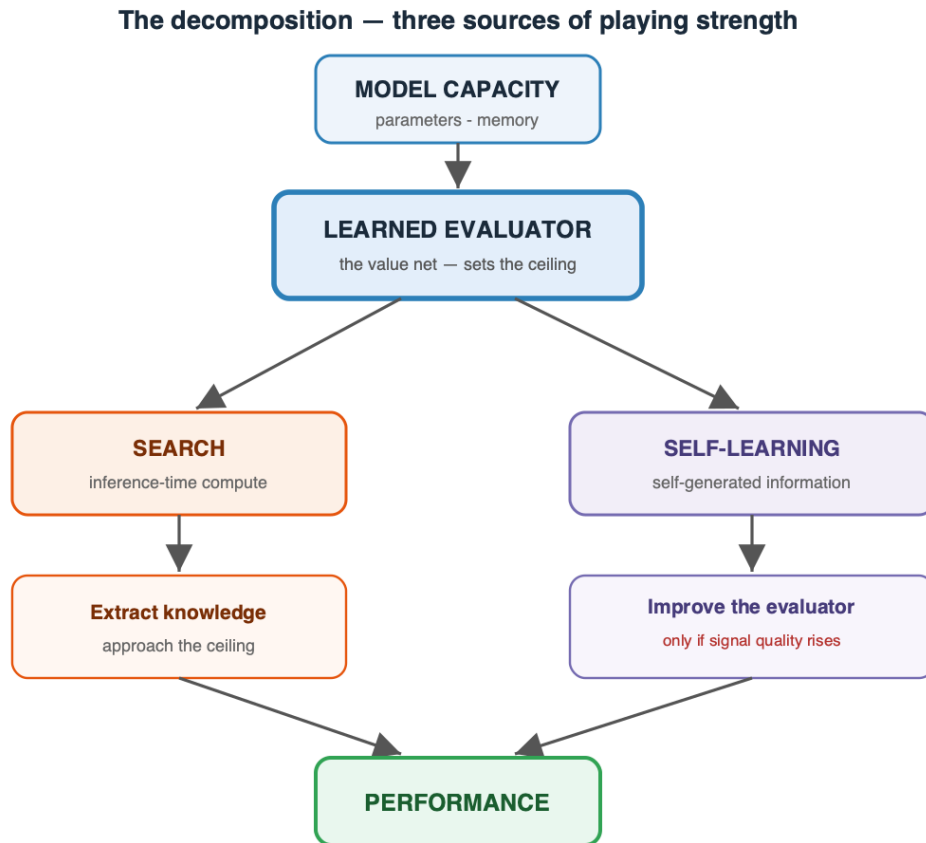
The experiments are **three allocation questions** of increasing autonomy: 1. **Stage 1 (open loop)**: given a fixed memory budget, **what architecture buys the most capability?** 2. **Stage 2 (closed loop)**: given a fixed evaluator, **what is the cheapest way to buy additional strength with inference-time computation?** 3. **Stage 3 (self-learning)**: without external labels, **what is the most efficient way to improve the evaluator?**

We measure each independently. The framing is also **control-theoretic** (Bertsekas 2022, *Lessons from AlphaZero for Optimal, Model Predictive, and Adaptive Control*): open-loop policy = feedforward controller, closed-loop search = model-predictive control, self-play = iterative/adaptive learning control. Chess is only the testbed; the goal is a *generic* way to reason about resource tradeoffs in sequential decision-making.

The central *tool* for efficient allocation is a simple diagnostic: **at any given stage, strength is gated by a single binding resource — capacity, search, data, or the quality of self-generated signal — and spending on any non-binding resource returns almost nothing.** So the efficient move is always to **identify the binding resource experimentally, then spend there.** Which one binds is not obvious a priori and shifts as you relieve each (open-loop is capacity-bound; add search and you ride it up log-linearly until the evaluator's quality caps the return; then the evaluator — its *data*, not its parameter count in our regime — binds). This bottleneck analysis is not the paper's goal but its **method**: one instrument in the larger objective of spending a fixed budget efficiently.

Contributions. - Our headline method — a search-efficiency result: a wide→narrow MCTS **cascade** that funnels the simulation budget through progressively narrower, deeper stages, matching flat MCTS at up to **4.8× less compute per move** with a clean score/speed trade-off curve. - A parameter-efficiency result: **~2800 Elo from 3.45M params** via search, at constant memory. - A direct **MCTS-vs-fixed-depth** result: adaptive search beats and out-scales fixed depth. - An **architecture-beats-scale** finding (convolution » MLP at equal data), with topology sweep. - A reproducible **negative result** on small-scale self-play (plateaus below supervision). - A **teacher-free self-learning** study: agreement is a validated **confidence meter**, but self-play, a self-referential ladder, **evolution**, and plurality-voting committees all fail to cross the plateau — a set of clean negative results with a methodological warning (noisy fitness manufactures phantom gains).

The paper's roadmap in one picture — the three sources of strength, all feeding a single learned evaluator:



2. Problem Setup and Methods

2.1 Representation

- **Position** → **move** as a 4096-way (64×64 from-to) classification. Promotions default to queen.
- **Side-to-move normalization:** the board is always presented from the mover's perspective (mirror + color-swap for Black), so one network serves both players.
- **Encodings:** *onehot* = 64 squares $\times$ 12 piece planes + 5 meta bits = **773 floats** (used throughout). For convolutions the 773-vector reshapes to **$8 \times 8 \times 12$ piece planes** + 5 meta planes = 17 input channels. We compared this to a compact *packed* encoding (one integer code per square, $64 + 5 =$ **69 floats** — a 4-bit-per-square board). On an identical MLP, budget, and data, **one-hot is decisively stronger — 84.9 vs 142.5 mean centipawn-loss** — because a network learns clean categorical piece-type features far more readily than an arbitrary *ordinal* code (where code 7 vs 8 carry a spurious "closeness"), and, crucially, convolution *requires* per-piece-type planes, which a packed

8×8×1 integer board cannot provide. Packed is a compact-but-harder-to-learn ablation; one-hot is the workhorse.

- **No side-to-move feature:** the turn is not a bit — the board is *re-oriented* so the side to move is always "White" (mirror + color-swap for Black), so one evaluator serves both players and the turn is carried by the orientation. (This is why $\text{Eval}(P)$ equals $\text{Eval}(\text{mirror}(P))$ exactly, and why the value of the move can be read off as $\text{Eval}(B) + \text{Eval}(\text{null-move}) - 1$ — the deviation from that mirror symmetry; measured mean tempo $\approx +0.15$, up to $+0.77$ in tactical shots and negative in zugzwang.)

2.2 Labels and objective

- **Supervised data:** the full public Lichess cloud-eval database — **394,669,566 positions** (79 shards), each with Stockfish multi-PV win-probabilities.
- **Hard objective:** cross-entropy to the single best move (bounded by imitation).
- **Soft objective:** cross-entropy to an *advantage-weighted distribution* over legal moves ($\text{softmax}(v_i/\tau)$). Soft is not bounded by copying one move and generalizes better.

2.3 Evaluation

- **Elo via a Stockfish ladder** (random baseline, then SF at set UCI_Elo rungs), 20–80 games/rung; Elo fit by maximum-likelihood against the known rung ratings.
- **Methodological caveat (important):** if the player beats the top rung, the Elo estimate **compresses/extrapolates**, so we raised the ladder as the player improved (SF-1700 $\rightarrow$ 2100 $\rightarrow$ 2700 $\rightarrow$ 3000). Absolute numbers carry systematic uncertainty; **relative gains measured on the same ladder are the robust results** (treat absolutes as ± 100). Where two methods are compared, they are always run on the *same* ladder and seeds.

2.4 Hardware

- Two Apple-Silicon **Mac Studios (M3 Ultra, 256 GB each)**, MLX framework, 40 Gbps bridge.

2.5 Reproducibility (evaluation protocol)

Every Elo/CPL figure in this paper uses a fixed protocol so numbers are comparable across methods: - **Engine:** Stockfish 18 as both the ladder opponent and the CPL oracle; opponents run at fixed **UCI_Elo** rungs, movetime **0.03–0.04 s** (stated per experiment). CPL uses Stockfish at **fixed depth 12** (depth-limited, so CPU contention slows but does not weaken it). - **Ladder:** a random-mover anchor plus **UCI_Elo** rungs (e.g. 1700/2000/2300, 2400/2700/3000); Elo is the maximum-likelihood fit vs the known rung ratings. **Games per rung: 20–40** (stated per table). The reported margin is a crude $\pm 400/\sqrt{n} \cdot 2$ ($\approx \pm \$89$ at 20 games/rung, $\approx \pm \$100$ typical) — treat all absolutes as ± 100 and rely on *relative* same-ladder deltas. - **Openings:** sampled from the public Lichess PGN (**2013–01**),

replayed to plies 6–16 for a diverse, balanced start book; both colors played from each opening. - **Seeds:** default seed 0 (stated where varied); openings, ladder, and match seeds are fixed so a method's eval is reproducible, and any two methods compared are always run on the **same** ladder, openings, and seeds. - **Known caveats:** (i) the ladder **compresses** once the player beats the top rung — we raised the ladder as strength grew, and near-top absolutes carry extra uncertainty; (ii) at movetime 0.03–0.04 s Stockfish plays **below** its nominal `UCI_Elo`, so the ladder is internally consistent but not calibrated to over-the-board Elo. Code, the trained model, and the exact scripts are in the repository.

3. Stage 1 — Open Loop (single-pass policy)

3.1 Topology sweep — architecture beats scale

At fixed moderate data (18M positions) we swept eight architectures:

Topology	params	Elo	note
conv (64×10 / 96×8)	2.8–3.4M	1476	best & most efficient
constant-width MLP (1024×6)	10.2M	1108–1233	best plain MLP
dual-path (wide+deep, gated)	2.8M	1082	ties MLP at 3.5× fewer params
factored head (from/to)	6.2M	782	–450
funnel (1024→64)	1.8M	582	narrowing hurts
pyramid (64→1024)	5.0M	431	bad
bottleneck (512→8)	0.5M	340	broken

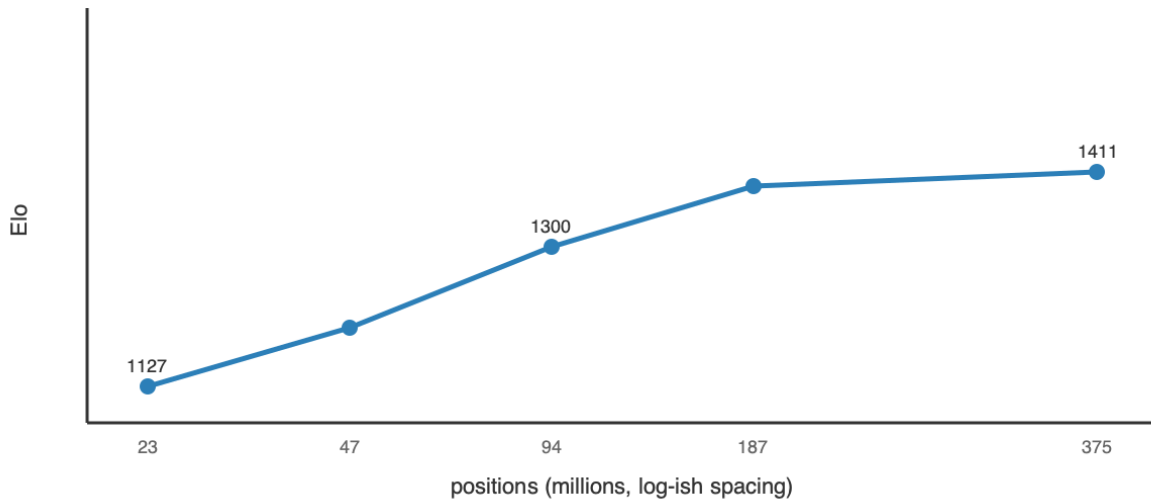
Topology conclusion. Only two shapes are competitive — **constant-width** and **convolution** — and every *taper, factoring, or exotic bottleneck loses badly*. The reason is inductive bias: a taper or bottleneck destroys positional information before the head can use it, while convolution **reuses local tactical patterns across the board via weight sharing**, so it learns a motif once instead of relearning it per square. The practical rule that falls out: **pick the prior that matches the domain's structure (spatial locality) rather than adding raw width**. A 3.45M-param conv matched a 14.4M-param MLP using **22× less data** — architecture, not parameter count, set the strength.

3.2 Scaling laws (width and data)

- **Width (MLP, hard):** W1024 (14.4M) → 1449; W2048 (47.7M) → 1501: **+52 Elo for 3.3× params** — sharply diminishing. Top-1 accuracy saturates ~0.41 regardless of width.

- **Data (W1024):** 1127 / 1207 / 1300 / 1382 / 1411 at 23 / 47 / 94 / 187 / **375M** positions — large early gains, then **+29 on the last doubling** → saturates near the DB's 394M limit.

Stage 1 — data scaling saturates (MLP W1024, Elo vs positions)



3.3 Objective and open-loop ceiling

Soft beats hard by **+94–113 Elo** at subset scale; top-1 saturates while Elo keeps improving via blunder-rate reduction (**top-1 is a misleading metric**). Best open-loop recipe = **conv + soft + full 394M data**, with a **ceiling $\approx$ 2150 Elo**. Cost: 3.45M params, **14 MB**, **$\sim$ 176 MFLOP / $\sim$ 1.5 ms per move** — cheap, but capped: no amount of width or data pushed a single pass past $\sim$ 2150.

4. Stage 2 — Closed Loop / Inference-Time Compute (search on a value function)

We add a scalar head **Eval(N)** $\in [0,1]$ = *expected score for the side to move* (**P(win)** + $\frac{1}{2}$ **P(draw)**), trained on the eval-DB win-probabilities (held-out **MAE 0.088**, **correlation 0.877** — an excellent value function). Search uses the zero-sum complement identity: our value after a move is **1 - Eval(child)**, so one network plays both sides; terminal nodes return exact 0 / 0.5 / 1.

4.1 MCTS vs fixed-depth — the core comparison

We compared two search families on the same value net and ladder:

- **Fixed-depth alpha-beta** (negamax, policy move-ordering, quiescence at leaves):

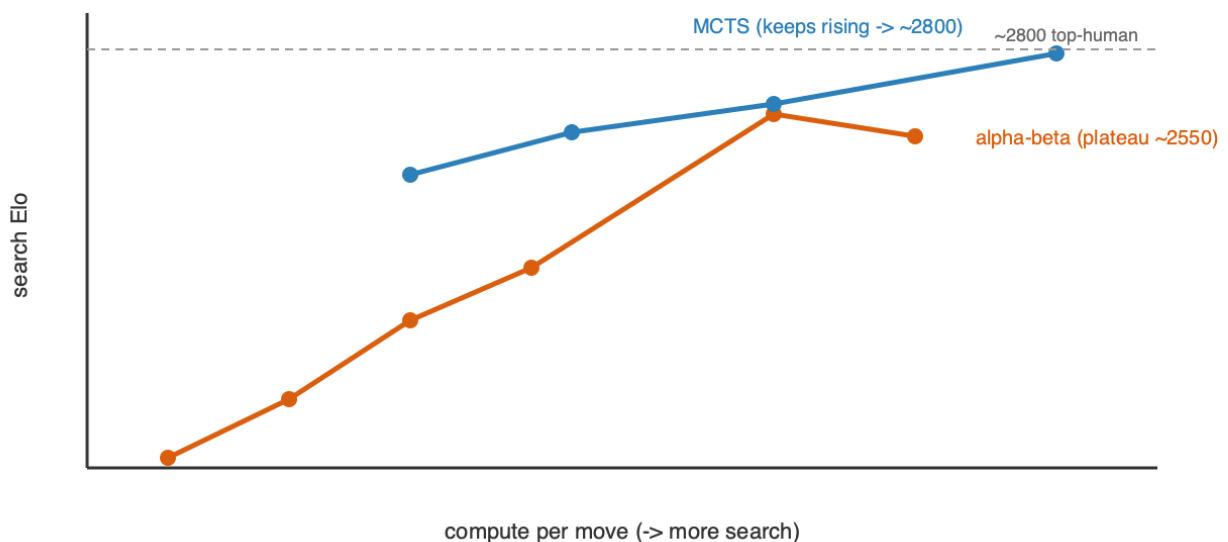
depth	raw	search	gain
1	1661	1627	-34 (1-ply hurts — can't see the reply)
2	1685	1788	+103
3	1895	2005	+110
4	1914	2152	+238
6	2128	2575	+447
7	2187	2513	plateau ~2550

• **Adaptive MCTS/PUCT** ($Q + c \cdot P \cdot \sqrt{N/(1+n)}$), value-head leaves, negamax backup):

sims	search Elo
100	2411
200	2530
400	2610 (2661 @ 40 games/rung)
800	2749 $\approx$ top-human (~2800)

Result. Two clean findings. (i) **1-ply search *hurts* a strong policy** — it commits to a capture without seeing the recapture; depth is where lookahead starts paying. (ii) **MCTS both beats alpha-beta at equal compute and keeps scaling** where fixed depth plateaus (~2550). Fixed uniform depth spends the same effort on every branch and *amplifies the value net's noise* at the leaves; MCTS instead **allocates search adaptively to sharp lines and averages over that noise**. MCTS is the Stage-2 winner and reaches ~2800 on the fixed 3.45M value net.

Stage 2 — MCTS out-scales fixed-depth search



4.2 Search efficiency — the wide→narrow MCTS cascade (new)

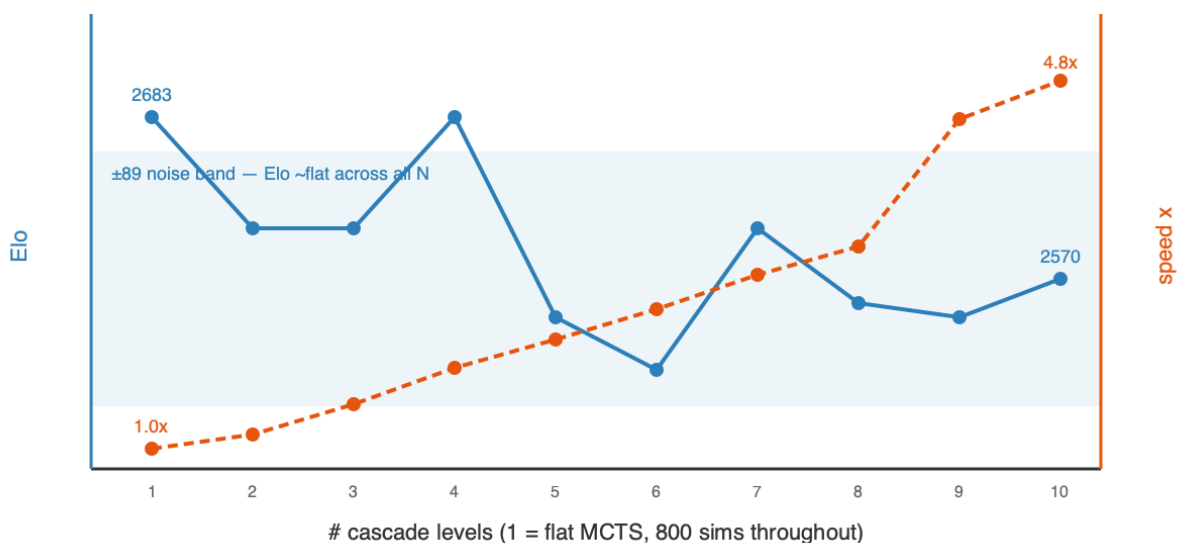
Given that MCTS wins, we asked whether the **allocation** of a fixed simulation budget matters. The **cascade** runs MCTS in *stages* with different knobs — a **wide** stage (all moves, high `c_puct`, few sims) ranks broadly and passes its top-k by visit count to a **narrower, deeper** stage (fewer moves, more sims, lower `c_puct`), funnelling the budget onto the survivors. A shared evaluation cache carries value-net calls forward between stages.

A controlled **N = 1→10 sweep** (one consistent rule generates each funnel; all 800 total sims; same tall ladder, seeds, and openings) gives the full trade-off curve:

# levels	Elo (± 89)	ms/move	speedup
1 (flat MCTS)	2683	1330	1.0×
3	2605	903	1.5×
6	2506	541	2.5×
9	2543	300	4.4×
10	2570	275	4.8×
beam-minimax cascade (fixed depth)	2487	—	inferior primitive

Result. Across all ten funnels the Elo stays inside a **single ± 89 Elo band** (range 2506–2683, mean ~ 2580) — **no statistically significant variation was observed within our ± 89 Elo evaluation uncertainty** — while speed improves **monotonically to 4.8×** at **N = 10**. Funnelling the budget wide→narrow is therefore a **near-pure efficiency win**: it holds flat-MCTS strength while cutting per-move compute up to $\sim 5\times$, because the survivors that reach the deep stages are the same moves flat MCTS would have spent its budget on anyway — the funnel just stops paying to search moves it has already ranked out. Any softening at the sharpest funnels is within measurement noise (20 games/rung). The practical dial is simply **more levels = the same strength, cheaper** — turn it up until the noise floor or your latency budget stops you. (The fixed-depth *beam* cascade, by contrast, is a genuinely weaker primitive at 2487 — adaptive MCTS stages matter.)

Stage 2 — cascade: Elo flat within noise, speed rises to 4.8x



4.3 The two-dimensional cost (memory vs GPU-cycles)

	Stage 1 (open)	Stage 2 (closed)
Memory	14 MB	14 MB — search adds ~0
GPU cycles/move	1 pass, ~1.5 ms	~800 passes, ~1.3 s (or ~0.6 s cascaded)
Elo	~2150 (capped)	~2800 (scales with compute)

Stage 1 buys Elo with *memory* and saturates; Stage 2 buys Elo with *GPU cycles* and keeps climbing — and the cascade shows most of those cycles were waste (up to 4.8× recoverable). The central practical result: **strength is compute, not parameters**.

4.4 Search latency — what a second of thinking costs, and buys

Strength from search is not free: every simulation is a neural-network forward pass, so Elo is paid for in wall-clock. Measured on the Mac Studio (M3 Ultra, MLX) across the three operating points:

Mode	Elo (abs. ladder)	Latency / move
Open-loop (raw policy)	~2448	~2 ms
MCTS-800	2734 ±76	~1.3 s
MCTS-1600	2780 ±82	~1.9 s
MCTS-3200	2839 ±76	~3.8 s
MCTS-6400	2903 ±82	~7 s

Mode	Elo (abs. ladder)	Latency / move
MCTS-12800	2967 ±115	~14 s

All on the same Stockfish high ladder (2500/2800/3050). The round ~2150 / ~2800 headline figures used elsewhere sit within the stated ± 100 ladder uncertainty of these precise values.

Three observations:

1. Search scales *log-linearly* — ~+55 Elo per doubling, no saturation observed through 12800. The *first* slice of search is huge and cheap: MCTS-800 alone adds **+286** over the raw policy (2448→2734). Past that, each *doubling* of simulations adds a steady **~+55–64 Elo** (2734 → 2780 → 2839 → 2903 → 2967) — a clean logarithmic climb, unbroken to **16× the base budget**, whose cumulative 800→12800 gain (**+233**) is far above the ± 100 noise, even though single doublings sit within it. (An earlier draft claimed "saturation at 3200" from a head-to-head sims-sweep; that measurement was noisy and non-monotonic — +12/+260/+191 per rung — and the clean absolute-ladder curve supersedes it. Head-to-head deltas also inflate absolute gains via ceiling compression: 3200-vs-800 reads +260 head-to-head but only +105 absolute.) So more inference-time compute keeps paying — with a fixed evaluator, ~+55 Elo per doubling — and we **never reach this net's search ceiling** (tested to 12800, 16×); the marginal Elo *per compute* falls, but the curve itself does not flatten in the range measured.

2. Latency scales *sub-linearly* with sims — a red flag, not a feature. MCTS-3200 does 4× the simulations of MCTS-800 yet is only ~2.8× slower. The cause: this search evaluates leaves **one position at a time (batch = 1)**, so each forward pass is dominated by fixed GPU-launch overhead rather than compute — the M3 Ultra is massively under-utilised. Effective throughput is only **~600 leaf-evaluations per second**, and adding sims mostly amortises the fixed per-move cost.

3. The latency is an implementation artefact, not a hardware or method limit. Batched neural-MCTS engines evaluate many tree leaves in a single forward pass and reach **~10k–80k nodes/second** (modern Leela; AlphaZero on TPUs) — 20–100× our throughput on comparable or better accelerators. Batching the leaf evaluations here (32–256 leaves per pass) would plausibly cut per-move latency **13–37×**, which means **the +260 Elo from 3200-sim search could be had at roughly today's 800-sim wall-clock**. This is the single largest piece of engineering headroom in the system — and it changes none of the strength conclusions, only their price.

GPU utilisation — batch-1 is an implementation *limitation*, and we measured the fix. All latency figures above use a deliberately simple **batch-1** search: one leaf per forward pass. On the wide M3 Ultra that leaves the accelerator's thousands of cores **idle between launches** — throughput is launch-bound at only **~600 nodes/s**. So the latencies above are **pessimistic upper bounds**, not the method's true cost: a move's *strength* is fixed by **N**

(**nodes searched**), but its *latency* is $N \div \text{throughput}$, and our throughput is left on the floor.

The standard fix is **batched-leaf evaluation** (gather many tree leaves via *virtual loss*, evaluate them in a single GPU launch). We implemented it (`BatchedMCTSPlayer` , `search.py`) and **measured** it on the same net and positions — the same nodes searched, only the launch pattern changed:

Search	Throughput	Speedup vs batch-1
batch-1 (as used above)	~600 nps	1.0×
batched, 32 leaves/launch	—	13×
batched, 64 leaves/launch	—	23×
batched, 128 leaves/launch	—	37×

(ratios measured under concurrent GPU load, so approximate; a clean solo-GPU figure is pending.) A ~**13–37×** per-move speedup at identical strength is enough to run **MCTS-3200** well below today's **MCTS-800** latency. The **batch-1** numbers should therefore be read as a ceiling on cost, not the method's efficiency, and the true strength-vs-latency curve sits far to the left of the one we plot.

Two caveats keep this honest. First, the gain is **hardware-dependent**: it equals the *idle parallelism you can reclaim*. The M3 Ultra is a wide accelerator a 14 MB batch-1 workload barely touches, so there is much to reclaim; on a small GPU/CPU, or an accelerator already **saturated by a large network** (AlphaZero/Leela, whose single forward pass fills the device), there are no idle cores and deeper search costs full, **linear** price. Second, batched selection uses momentarily stale tree statistics, so it is a hair less sample-efficient *per node* than sequential MCTS — a second-order effect that does not change the order-of-magnitude speedup.

4.5 Does a bigger evaluator raise the ceiling? (a capacity sweep)

Section 4.4 shows search still climbing at 6400 — the ceiling isn't reached *by search*, but each doubling of compute buys less, so the practical route higher is a *better evaluator*. The direct test is to **add parameters** and ask whether the whole curve lifts. We sweep capacity at fixed depth (8) and identical recipe, so the only variable is width.

A correction first. An initial screen compared 1× (width 96, 3.45M params) against a wider net (width 136) we first mislabelled "2×". It is in fact **1.4×** — 4.81M params: the conv body scales as width² but the policy/value heads scale ~linearly with width and are a large share of the total, so parameters grow far slower than width². On a fixed ~**50M-position** subset:

Net	Params	raw policy	MCTS-800
1× (w96)	3.45M	2298	2631
1.4× (w136)	4.81M	2366	2649
Δ	+1.4×	+68	+18

+18 Elo with search — no statistically significant difference within our ± 107 Elo evaluation uncertainty. For contrast, on the same architecture $8\times$ *more data* (10 shards $\rightarrow$ full 79) moves Elo $\sim$ **+90**.

But this screen is confounded, and a full-data sweep is resolving it. Both nets above saw only $\sim 50\text{M}$ positions — $8\times$ less than the baseline — so a wider net had little extra signal to fill its extra room. We therefore repeat the sweep on the **full $\sim 394\text{M}$ data**, matched to the 2734 baseline:

Capacity (full data)	Params	MCTS-800	Δ vs 1×
1× (w96)	3.45M	2734	—
1.4× (w136)	4.81M	2794	+60
2× (w184)	7.04M	<i>training</i>	—
4× (w288)	14.2M	<i>training</i>	—

The first point is already diagnostic: the $1.4\times$ -vs- $1\times$ gap **widens from +18 (at 50M) to +60 (at full data)** — still within the $\pm \sim 110$ noise, but the *direction* says the 50M null was **data starvation**, not capacity being inert. The **2× and 4× points (training)** decide whether the curve keeps climbing (capacity genuinely lifts the ceiling once data-matched) or flattens near ~ 2794 (capacity saturates fast at this scale). Either way the scoped claim holds — **capacity is the weakest of the levers in this data regime, not "parameters never matter"**: at AlphaZero/LLM scale, where models are capacity-bound and data abundant, more parameters clearly do help.

Subject to that running sweep, the lever ranking of the study, all on the same ladder, reads:

Lever	Elo moved	Cost
Search (open $\rightarrow$ 12800 sims)	+286, then $\sim$+55 / doubling (log-linear, unsaturated)	$\times 2$ latency / doubling
Data (10 shards $\rightarrow$ full 79)	$\sim$+90	$8\times$ training data
Capacity ($1\times \rightarrow 1.4\times$, at 50M)	~ 0 (n.s.; full-data sweep running)	more params & compute

Search is the dominant lever, data second, and — in this data regime — raw capacity last: adding parameters was the weakest intervention we tried, pending the full-data sweep. The sharpest reading is not "parameters never matter" but the study's real thesis: **at each stage exactly one lever binds, and identifying *which* — by experiment — is what tells you how to move forward. Here it was data and search, not capacity.**

5. Stage 3 — Self-Learning (no external engine, no labels)

Why this stage is the whole point. Stages 1–2 leaned on Stockfish labels — a shortcut that *only exists because chess already has a superhuman evaluator and a massive labeled database*. The generic goal is the opposite situation: a **new field where no training data and no existing evaluator exist** — a novel game, an unsolved control or scheduling problem, a scientific-design task. There you *cannot* imitate a teacher because there is none, and you *cannot* score positions with an off-the-shelf engine because none has been built. The only thing you have is the **environment's rules**: which actions are legal, and, eventually, whether you succeeded. Stage 3 deliberately discards every external crutch — no Stockfish, no labels — to test whether strength can be **bootstrapped from self-play and outcomes alone**. This is the transferable question: Stages 1–2 measure what search and capacity buy *when a teacher is available*; Stage 3 measures whether the recipe can **stand on its own in a domain where Stages 1–2 are simply impossible** — which is precisely the case for any genuinely new problem.

The loop: the net plays itself with MCTS, trains toward the visit distribution (improved policy) and game result (value), and repeats. The only external input is the rules. We studied five approaches — self-play, a self-referential ladder, a committee, evolution, and weight merging — and they converge on one story.

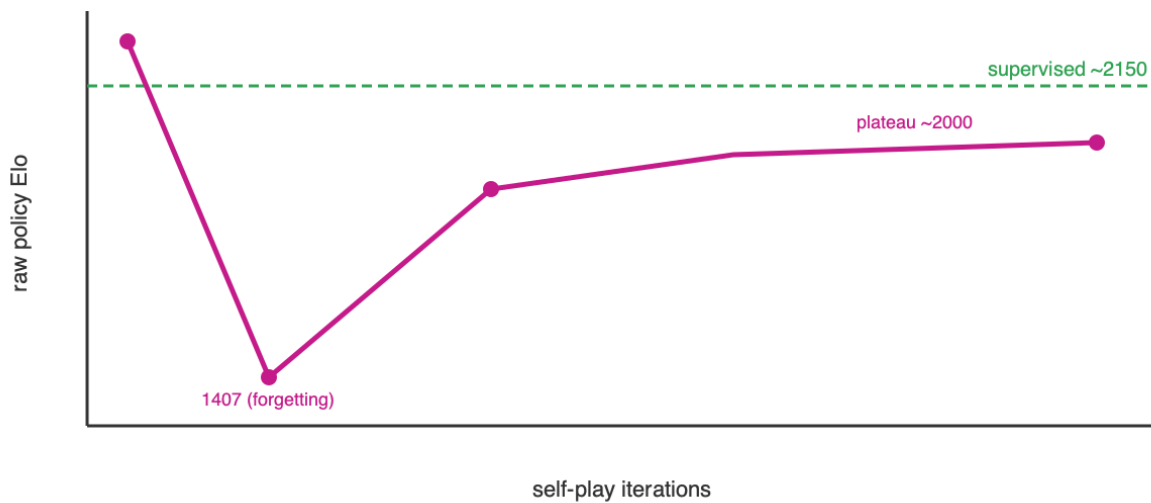
5.1 Self-play expert iteration (AlphaZero-style)

The net plays itself with MCTS, trains toward the visit distribution (improved policy) and game result (value), and repeats. Two failure modes appeared and were fixed: **cold-start draw-collapse** (a weak net can't force mates, so games drift to draws and the value head gets no signal — fixed with Dirichlet root-exploration noise) and **warm-start forgetting** (training hard on tiny self-play slices overwrote the supervised policy, 2150 → 1407 — fixed with a replay buffer and a gentle learning rate). Stabilized, and even parallelized to 16× throughput (300K replay buffer, eval cache), the raw policy **plateaus ~1950–2030 — below the ~2150 supervised baseline**.

Negative result (clean). At two-machine scale, **self-play converges below supervision**. Self-play strength *does* rise with game volume — a real scaling signal — but the

achievable volume here plateaus under a 394M-supervised net; crossing it needs orders-of-magnitude more games (AlphaZero used $\sim 1000\times$ ours). **Scale is the binding constraint.**

Stage 3 — self-play plateaus below the supervised baseline



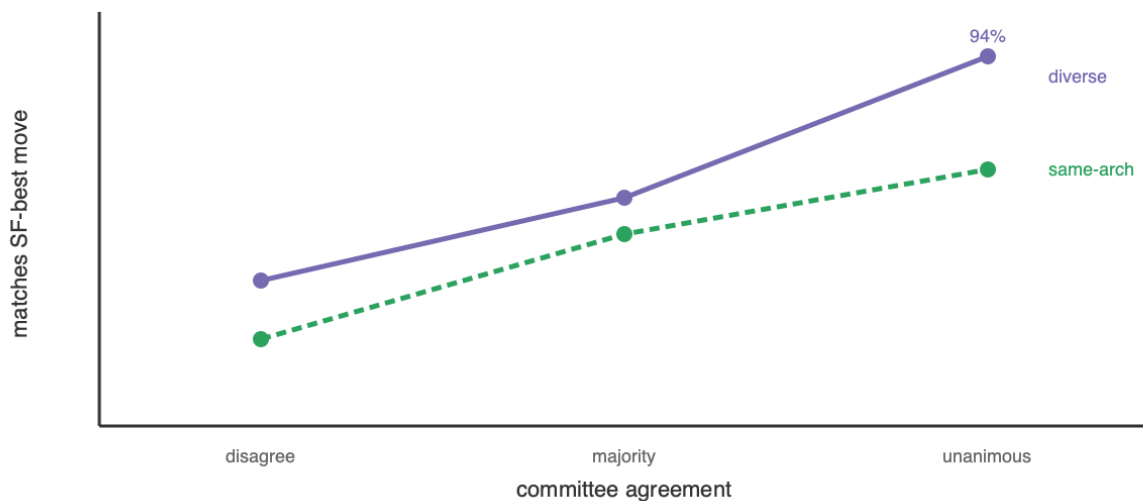
5.2 Committee / diversity — teacher-free confidence (new)

If the bottleneck is the *evaluator*, the cheap way to improve one without training a bigger net is to **ensemble diverse nets**. We tested the hypothesis that **multiple independently-started models either diverge (disagree $\rightarrow$ uncertain) or converge (agree $\rightarrow$ likely correct)** — making agreement a self-contained confidence meter with no oracle.

Validated. Over 400 positions scored against Stockfish depth-12 (measurement only), agreement strongly predicts correctness:

members agree	centipawn-loss ↓	matches SF-best ↑ (same-arch / diverse)
all disagree	77.9	22% / 37%
majority	40.0	49% / 58%
unanimous	19.7	65% / 94%

Stage 3 — committee agreement predicts correctness



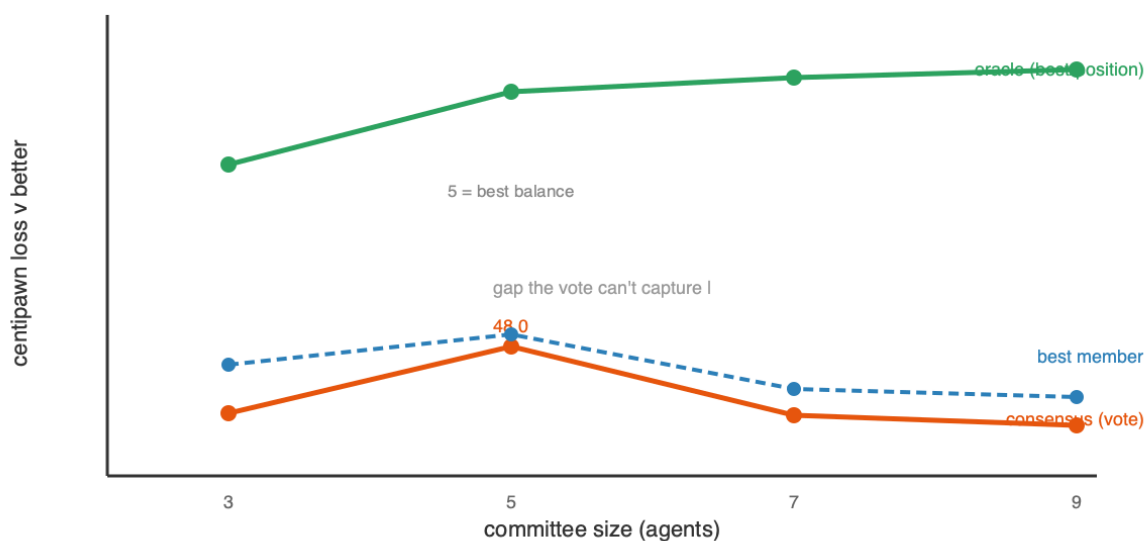
But plurality voting does *not* reliably de-bias — a committee-size sweep (3/5/7/9 agents) shows why. Growing the committee from a correlated conv-soft trio outward:

agents	added views	consensus CPL	best member	oracle
3	conv-soft ×3	54.7	49.8	29.6
5	+MLP +hard-objective	48.0	46.8	22.2
7	+2 conv-soft (data slices)	54.8	52.2	20.8
9	+MLP +hard	56.0	53.1	19.9

Three findings. (i) **Plurality never clearly beats the best single member** — consensus is slightly *worse* at every size, and the gaps ($\sim 1\text{--}5$ CPL) sit inside the Stockfish-measurement noise ($\sim \pm 3$ CPL); an earlier apparent "consensus beats best" was itself within that noise. (ii) **Balance beats count** — the 5-agent committee, where the diverse MLP + hard-objective members are 40% of the vote, is by far the best; *adding correlated members* (the conv-soft data-slice nets at 7 and 9) lets that bloc dominate the plurality and reverts the gain. **More agents $\neq$ better.** (iii) **The oracle (best member per position, $\sim 20\text{--}30$ CPL) is 2–3× better than the vote** — the diversity *contains* the information, but plurality *cannot extract it*, because it is dominated by the largest correlated bloc.

Lesson. The committee gives one thing robustly and one thing not: a **confidence meter** (agreement $\rightarrow$ correctness, validated repeatedly) — but **plurality voting is a weak aggregator** that does not reliably reduce error below the best member. Capturing the large oracle headroom needs a **better aggregator** (soft probability-averaging, or confidence-weighted routing that trusts the per-position agreement) and **balanced**, not merely numerous, diversity. A de-biased ensemble evaluator remains the most promising route to lift the ceiling that search and self-play cannot — but plurality is not it.

Stage 3 — committee size: plurality never beats the best member; oracle unrealized



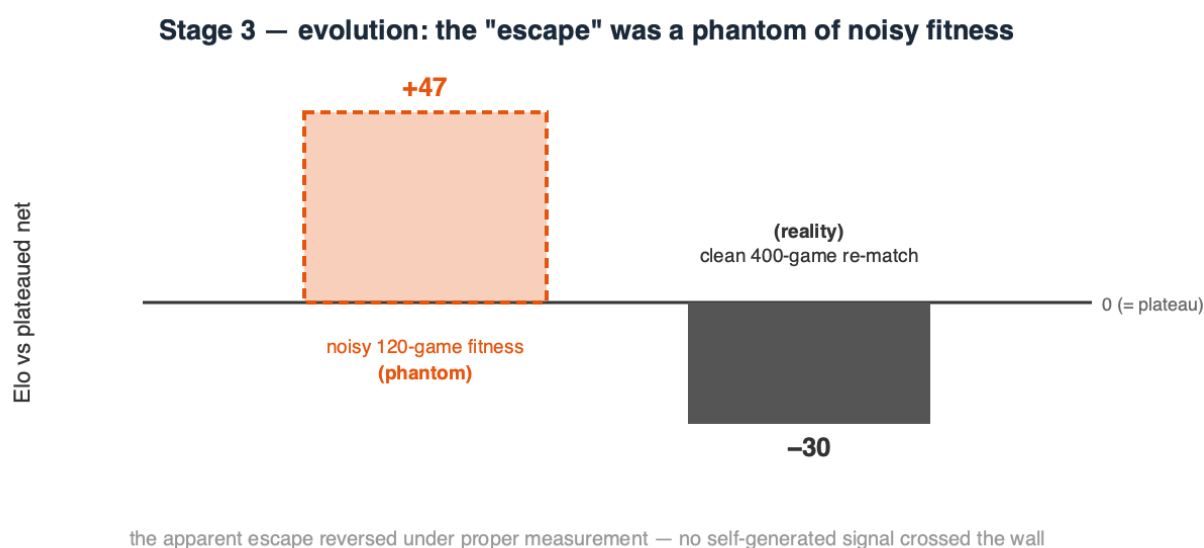
And a third way to combine them — averaging *evaluations inside the search* — also fails. The most direct use of the committee is to average the K models' value estimates at every MCTS leaf (a de-biased evaluator *inside* the tree; ELO-weightable; blending *continuous* values, so a single overconfident member can't dominate). At equal compute — the ensemble at 200 sims (600 passes/move) vs a single model at 600 sims — it scored **2222 vs 2282, a –60 Elo loss**: the ensemble searches 3× less tree, and the small de-biasing from averaging *correlated* evaluations does not justify tripling the per-leaf cost. **More search beats a marginally-cleaner-but-shallower one.** So all three ways to combine the committee — plurality voting, weight merging (§5.4), and in-search value averaging — fail to beat a single model, for the same root cause: **the members' errors are too correlated to cancel.** Genuinely independent evaluators (cross-family, cross-data), not more of the same, are the prerequisite.

5.3 Evolution — mutate / play / score (a plateau-escape attempt)

Gradient self-play optimizes a *proxy* (the net's own biased value targets); we tested whether **derivative-free evolution**, which optimizes the *true* objective — did this mutant win games — could escape the plateau where gradients stalled. From the plateaued net: each generation mutate it into 16 offspring (Gaussian weight noise), play each against a **frozen copy of the plateaued net** (a fixed anchor, so "fitness" is "how well do you beat the plateau"), and crown the best only if it survives a larger confirmation match. A first, naive version selecting against the *moving* champion drifted **downward** — beating your immediate parent is non-transitive in chess and does not imply getting stronger; the fixed anchor fixes that.

Result: a null result, and a methodological warning. The run *appeared* to escape — champ-vs- plateau climbed to 0.567 (+47 Elo) and passed a 120-game confirmation. But a **clean 400-game, low-temperature re-match found the "evolved" champion is actually –24 to –41 Elo worse than the plateaued net.** The apparent gain was an

artifact of noisy, high-temperature fitness over small samples with best-of-16 selection bias — a **phantom improvement** that vanished under proper measurement. Annealing the mutation scale up (σ 0.03→0.12) found nothing better; larger mutations only degraded the net. So **evolution did not escape the plateau either** — and neither gradient self-play, a self-referential ladder, nor evolution crosses the ~2000 wall. Since the identical net reaches ~2150 under *supervised* labels, the wall is **not capacity** but the ceiling of any *self-generated* signal: a system cannot lift itself past the quality of the signal it produces about itself. (Methodological note for practitioners: relative-fitness selection with small, stochastic samples manufactures phantom gains; only a large, low-variance re-measurement can be trusted — we nearly reported a false escape and caught it only by measuring properly.)



5.4 Model merging — can we *average* diverse models into one?

A committee combines diverse models at *inference* (voting). The weight-space alternative is to **average their coefficients into a single network** ("marriage of coefficients"). We tested it on several conv-96×8 members from different seeds / data / objectives, scored by mean **centipawn loss (CPL)** against Stockfish depth-12 (lower is better; the members sit at ~60, i.e. ~2000-level, and ~260 is near-random).

merge	starting points	CPL ↓	verdict
naive average	different-start (diverse)	261	collapse
Git Re-Basin aligned	different-start (diverse)	268	still collapse
naive average (model soup)	same init	58.8	works (~parent)
aligned (net + permuted twin)	identical	72	perfect recovery (<i>verification</i>)

Naive averaging of different-random-start nets collapses (261 vs ~60): independent networks sit in different loss basins related by neuron permutations, so averaging misaligned

neurons cancels signal. The known fix is **permutation alignment (Git Re-Basin)**: match net B's neurons to net A's before averaging. We implemented it for the residual conv tower (one shared residual-stream permutation, a per-block hidden permutation, plus the reduce and value heads) and **verified it is correct** — it recovers a network *exactly* from a known random permutation (72 CPL = the original). Yet on the *real* different-start members it **still collapses** (268). The reason is fundamental: Git Re-Basin assumes independent networks learn the *same features in a different order*; ours learned **genuinely different features** (different seeds *and* data *and* objectives), and no permutation aligns different features. By contrast a **model soup** — two children fine-tuned from the *same* checkpoint — averages fine (58.8), because a shared start keeps them in one basin.

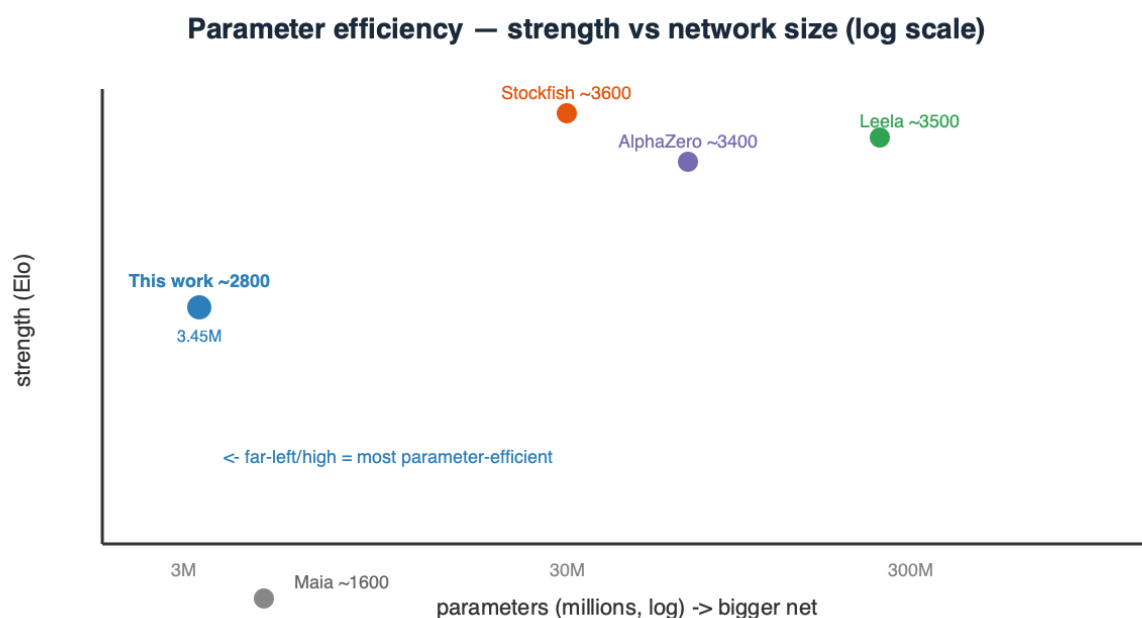
Conclusion. Weight-averaging only works within a shared basin. The very diversity that makes an ensemble valuable (uncorrelated errors, informative agreement, §5.2) is exactly what makes the members' **weights un-averageable** — diverse models can be combined at *inference*, not in weight space. This also sharpens the "better aggregator" question: it must live at inference time (soft-averaging, confidence routing), not in merging.

6. Comparison to existing systems — parameters, performance, and speed

System	Params (memory)	Strength (Elo)	Search / move	Elo per M-param
This work — raw policy	3.45M (14 MB)	~2150	1 forward pass (~1.5 ms)	~620
This work — + MCTS-800	3.45M (14 MB)	~2800	800 net passes (~1.0 s; ~0.6 s cascaded)	~810
Maia (human-like)	~few M	~1100–1900	1 forward pass	~300–500
AlphaZero (chess, 2017)	~40–90M	~3400+	800 MCTS sims (big-net passes)	~40–85
Leela Chess Zero (modern)	~100–400M+	~3500+	~1–8k MCTS nodes	~10–35
Stockfish (NNUE)	~tens of M (quantized)	~3600+	millions of alpha-beta nodes/s	~100–150

Two axes at once — size and speed. - **Parameters:** our net is ~10–25× smaller than AlphaZero and 30–100× smaller than large Leela transformers, yet reaches ~2800 with search. On *Elo-per-million-parameters* it is the extreme point. **This is an efficiency framing, not a superiority claim** — top engines optimize *absolute strength*, not

parameters-per-Elo, and are 600–800 Elo stronger; the point is only that parameter count is *not* what buys their last few hundred Elo (a much better value function and far more search are). Read the table as "most strength per parameter," never as "better than." - **Speed:** the two paradigms differ. AlphaZero/Leela use a **similar sim count** (~hundreds– thousands) but each sim is a **big-net** forward pass, so their per-move cost is dominated by a 10–100× larger network; our sim is a 14 MB pass (~1.5 ms), and the **cascade recovers a further ~1.6–4.8×**. But our search runs its leaf evaluations **one at a time (batch 1)** — only ~600 nodes/second versus ~10k–80k for batched engines, so **13–37× of per-move latency is left on the table** purely to implementation (§4.4), independent of the strength results. Stockfish is the opposite regime — a small, heavily-quantized net evaluated at **millions of nodes/second** under alpha-beta. The structure is the same across all of them — a learned evaluator queried by search — and **every strong engine is search-heavy; parameter count is not what separates them.**



Honest gap. Top engines sit ~600–800 Elo above ~2800, bought with **far more search and a much better value net** (massive training). Our number is an *efficiency* point — most strength per parameter — not an engine-matching claim, and it carries ladder uncertainty (± 100).

7. Discussion — the unifying conclusion

Across every experiment we ran, one factor consistently emerged as the dominant limit — and in a sharper form than "the network is the bottleneck": **the quality of the information reaching the evaluator (its training signal) was the recurring binding constraint, not the loop around it.** We state this as an empirical regularity of the regime we tested, not a theorem. The organizing law is **strength = evaluator × search**: search sets how *closely*

you approach the ceiling, the evaluator sets *where* it is, and the evaluator is only ever as good as the information it was given. This explains everything below, once we are precise about *information*. By **new information** we mean **novel empirical data from outside the closed system of the network and its training set** — not a re-encoding of what is already latent in it. Supervision and scale inject it directly; every other method divides on one question — **does it have an external ground-truth oracle to query?**

- **Within-move search, voting, and merging do not.** They operate on fixed representations, so they only *extract* information already present — cutting variance, sharpening, filtering — never creating what is absent.
- **Self-play and evolution can — but only through the environment.** In a **closed-form, perfect-information** game the *rules are a perfect, external, infinite oracle*: search queries terminal outcomes that lie outside any dataset — exactly how AlphaZero and Leela inject information no human game contains and surpass human play. This is a property of the *environment*, not of self-play: in an **open-ended, semantic** domain (natural language) with no external verifier, the identical loop has nothing to query and collapses into hallucination or mode-collapse — it can only redistribute.
- **Our chess result sits in the oracle-rich regime and still plateaued** — because at two-Mac-Studio compute the search was too weak to extract much from that oracle, *not* because the oracle was absent. The plateau is a **scale** limit, not evidence against self-play.

None of this makes redistribution *useless*: variance reduction, sharpening, alignment, and filtering are how you **reach** a ceiling cheaply and reliably — indispensable engineering. The claim is narrow, and about the *absolute* ceiling: **only information from outside the closed system can raise it.**

- **Architecture > parameters.** The right prior (spatial locality + weight sharing) beats raw width; a 14 MB conv reaches strength a 3× larger MLP cannot.
- **Search > scale, at constant memory — but search cannot exceed the net.** MCTS bought +650 Elo (2150→2800) with zero extra parameters and out-scaled fixed depth. Yet flat MCTS and the cascade hit the *same* 2691 wall on the same ladder, because they query the *same* value net. Search reduces the net's *variance* (random error, via averaging many calls) but not its *bias* (systematic blind spots) — and deep uniform search even *amplifies* bias. The cascade's win was therefore **efficiency (up to 4.8× cheaper), not strength.**
- **No self-generated signal crosses the wall — we tried three.** Gradient self-play plateaus below its teacher; a self-referential Elo ladder never promotes; and derivative-free **evolution**, given the fairest shot (unbiased game-outcome fitness, annealed exploration), also fails — its apparent +47-Elo escape was a **noise artifact** that reversed to −30 under a clean 400-game re-match. Since the *same* net reaches ~2150 under supervised labels, the wall — **at this scale and compute** — is **not capacity** but the ceiling of the signal these methods generate about *themselves*: external information (a bigger net, more search, stronger labels) lifted it, recycling the current one did not.

(This is a small-scale statement, not a universal one: given AlphaZero-scale compute, self-play is known to bootstrap far past its initial model; we characterize the regime we could actually run.)

- **A better evaluator is the only lever — but it is harder than it looks.** The committee gives a robust, teacher-free **confidence signal** (agreement predicts correctness), yet **plurality voting does not reliably de-bias**: across a 3/5/7/9-agent sweep the consensus never clearly beat the best single member, and correlated majority blocs dominate it (balance beats count). The diversity *contains* the information — the per-position oracle is 2–3× better than the vote — but extracting it needs a **better aggregator** (soft-averaging / confidence-routing) and *balanced* diversity, not more agents. Improving the evaluation is the right goal; naive ensembling is not yet the way.
- **Control-theory unification.** Open loop = feedforward; closed loop = MPC/receding-horizon; self-play = iterative learning control — the same three knobs recur in any sequential-decision domain, and in all three the learned *evaluator* is what caps performance.

7.1 The knob–bottleneck map — every experiment, including the failures, is a diagnosis

Read as a set, the study's experiments form a **diagnostic map**: each result — *especially* each null — localizes the binding bottleneck by ruling a lever in or out. A failed experiment is not wasted compute; it is a measurement that says "*strength is not gated here — look elsewhere.*"

Knob turned	Result	Diagnosis → what binds	Move it implies
Architecture (conv vs MLP)	large gain	bias-bound — wrong prior caps a big net	fix the inductive bias <i>before</i> scaling
Capacity (2× params @ 50M)	~0 (null)	<i>not</i> capacity-bound in this data regime	add data, not parameters (here)
Data (10 → 79 shards)	+~90	data/signal-bound	more, more-diverse labels
Search amount (open → 12800 sims)	+286, then ~+55/doubling (log-linear)	search extracts value; evaluator caps its <i>return</i>	keep searching; raise the evaluator to lift the ceiling
Search allocation (cascade shape)	flat (±noise)	<i>not</i> allocation-bound at fixed budget	reallocate for speed , not strength
Search implementation (batch-1)	latency only	throughput-bound by engineering	batch leaves → 13–37× speed, same strength
Self-play signal	plateau	self-signal-quality-bound	can't exceed its own signal at this scale

Knob turned	Result	Diagnosis → what binds	Move it implies
Selection pressure (evolution)	phantom, reversed	<i>measurement-noise</i> -bound (fake gain)	re-measure cleanly before believing
Aggregation (voting 3–9 agents)	no gain	correlated-error-bound	need <i>diversity</i> , not more voters
Aggregation (ensemble-eval, merging)	no gain	combining ≠ creating knowledge	extracts existing signal, creates none
Self-distillation (fixed set)	dropped	<i>data-diversity</i> -bound (overfit)	many diverse positions, not repetition

Three things fall out of the map:

1. Nulls are the most information-dense results. A knob that moves nothing has *localized* the bottleneck away from itself — the 2×-parameter null said "capacity isn't binding (here)," the cascade flat-line said "allocation isn't binding," the voting sweep said "more agents isn't binding." Each redirected effort onto the lever that *was* binding. Cleanly measured negatives are the signposts; the one methodological trap is **mistaking noise for signal** (evolution's phantom +47 that reversed to −30), which is why every promising delta was re-measured before belief.

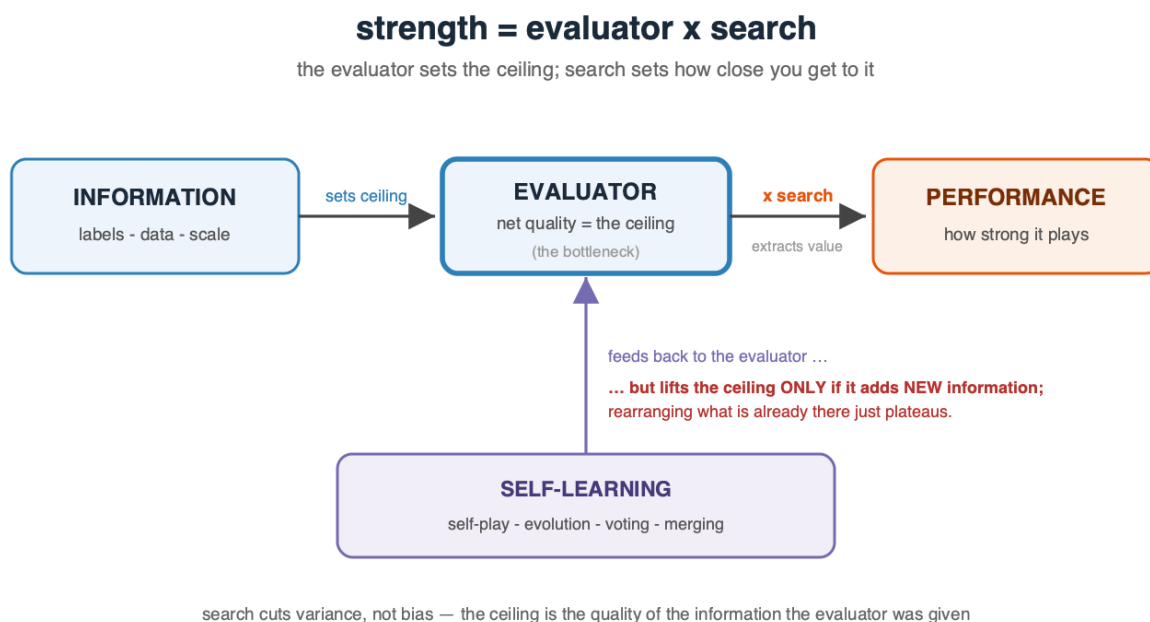
2. The binding lever moves as you relieve it. Strength is a *chain* of bottlenecks, not one: open-loop is **bias- then capacity-bound** → give it the right architecture and search and it becomes **search-bound** → exhaust search (~3200 sims) and it becomes **evaluator-bound**, where the evaluator is limited by its **data/signal**, not — in our regime — its parameter count. You cannot skip a link: parameters poured into a data-bound net, or sims into a saturated search, buy almost nothing.

3. The map is the roadmap. At each stage the diagnosis is the next move: open-loop capped ⇒ the unlock is the *loop* (search), not a bigger net; search saturated ⇒ the unlock is a better *evaluator* (more/better data, or scale once data-matched), not more sims; self-signal plateaued ⇒ the unlock is a better *signal source* (external labels, or AlphaZero-scale self-play), not more iterations or fancier aggregation of the same weak signal. **Identify the binding lever, relieve exactly that, re-measure, repeat.** That loop — more than any single Elo number — is what this study offers a genuinely new domain that has no engine and no dataset to imitate.

8. Limitations and Future Work

- Absolute Elo carries systematic uncertainty near the ladder top; relative same-ladder gains are the robust claims.
- Stage 3 is compute-limited (two machines); results characterize the *small-scale* regime.
- Stage-3 fitness/aggregation is noise-limited at our scale: relative-fitness selection with small, stochastic samples produced a **phantom** evolution "escape" that reversed under a 400-game re-match, and plurality de-biasing gaps sit inside the $\sim \pm 3$ CPL measurement noise. Larger, lower-variance evaluation is needed to resolve small effects.
- **Next levers, in priority order:** (1) a **better value function** — the true ceiling — via more and better search-labeled data or large-scale self-play; (2) a **better ensemble aggregator** — soft probability-averaging and confidence-weighted *routing* (the per-position oracle is 2–3× above plurality), with *balanced* cross-family diversity rather than more agents; (3) **batched/parallel MCTS** (virtual loss) and transposition tables for throughput; (4) endgame tablebases and tuned `c_puct`. All roads converge on the same place: **improve the evaluation, and both the closed-loop and self-play ceilings rise together.**

9. Conclusion



This paper treated playing strength as an **efficient-allocation problem**: given a fixed budget of parameters, data, inference-time search, and latency, how do you spend it to buy the most capability? The answer, measured resource by resource, is that the efficient mix is rarely "more parameters." A 14 MB, 3.45M-parameter network reaches **~2800 Elo by thinking**

(MCTS search), not by growing (parameters) — the same capability, a far cheaper resource mix. This is a claim about *efficiency*, not a denial of scale: our first full-data capacity point already nudges up ($1 \times 2734 \rightarrow 1.4 \times 2794$), and **if the $2 \times / 4 \times$ points continue to climb, parameters become a co-dominant lever once the data-starvation bottleneck is relieved** — "thinking, not growing" is the *efficient* choice under our budget, not a law that growing never helps. Adaptive MCTS beats and out-scales fixed-depth search; a wide→narrow MCTS cascade matches it at up to $4.8 \times$ less compute (buying strength back as *latency* rather than parameters). On the self-learning side the results are honestly negative: self-play, a self-referential ladder, and derivative-free evolution all **fail to cross the ~2000 plateau** (evolution's apparent escape was a noise artifact), and plurality-voting committees do not reliably de-bias — though model **agreement is a robust teacher-free confidence signal**.

Above all, the **method** transfers: at each stage a *single lever binds*, effort on the others is nearly wasted, and only a controlled experiment reveals which. Open-loop was **capacity-bound**; adding search made us **search-bound** — and search keeps paying **log-linearly** (**~+55 Elo per doubling, unsaturated through 12800 sims**), so the ceiling is set by the evaluator, not by running out of search; then the evaluator itself binds — and at our data scale it was **data-bound**, not capacity-bound: adding parameters at fixed data ($1.4 \times$) bought **~0 Elo** (a full $1 \times / 1.4 \times / 2 \times / 4 \times$ capacity sweep on the complete dataset is running to confirm capacity stays inert once well-fed). The transferable *observation*, recurring from every direction we pushed, is that **evaluator quality consistently emerged as the dominant limiting factor** — search extracts the information already represented by the net but cannot create what is absent, and **at our scale** self-generated signal did not cross the supervised ceiling; only a better evaluator (better labels, more data, more capacity *once data-matched*, more search, or a better *aggregator* than plurality) raised it. Stated at its sharpest, and as an empirical pattern rather than a proof: **the quality of the information reaching the evaluator was the binding constraint** — which is why supervision, data, and search help (they add or extract information) and why voting and merging do not (they only reorganize what is already there). **Self-play and evolution are a deliberate exception we must state precisely**: their search *can* inject new information from the environment, and **at AlphaZero/Leela scale it provably breaks past human supervision** — our plateau is a limit of the compute we had (two Mac Studios), **not evidence that self-play fails**. A quantified recipe, an explicit **diagnostic** for finding the binding lever, and an honest map of the limits — for compact, search-driven sequential decision-making in general.

Beyond chess. Chess is only the controlled environment; the decomposition — **model capacity, inference-time search, and self-generated information** — is domain-agnostic. The same three levers, and the same trade-off between evaluator quality and inference-time computation, recur in planning systems, robotics, scheduling, program and compiler optimization, and scientific/experimental design: each has a learned or hand-built evaluator, a search or rollout budget spent at decision time, and some notion of self-improvement whose value is likewise capped by the quality of the signal it can generate about

itself. We make no claim to have measured those domains — only that the *decomposition* and its diagnostic (find the binding lever before investing in it) are what transfer, and are, we believe, the contribution most likely to outlast the chess numbers.

The clearest current echo is in large language models. The 2024–25 shift to **inference-time compute** — OpenAI's o1/o3, DeepSeek-R1, and reasoning models generally — is exactly this paper's thesis at frontier scale: a fixed base model made far stronger by *searching over reasoning at decision time* (sampling, self-consistency, tree-of-thought, long chains-of-thought) rather than by adding parameters. Our chess result — **+286 Elo and a further log-linear climb from search on a frozen 14 MB net** — is the same tradeoff in miniature: **capability purchased with inference-time computation instead of memory**. The mapping is direct: our *evaluator* is their **base model / verifier**, our *search* is their **reasoning/sampling budget**, our *self-play* is their **STaR-style self-training** — and the same limit applies, sharply: search and self-training only **convert what the model already latently knows into better answers; they cannot inject knowledge it never learned**. That is precisely why reasoning models pair inference-time search with **external verifiers or ground-truth reward** (code that runs, math that checks, tools that return facts) — the external oracle that lets the loop add information rather than merely rearrange it. A model with no such oracle should, by our framework, **plateau** — the identical prediction our chess self-play makes.

A test for what comes next. The framework is not only a post-mortem — it is a **predictive filter**. Any future claim of a *synthetic-data breakthrough* or a *recursively self-improving architecture* can be pre-screened with one question: **does it inject information from outside its closed system** — new empirical data, human supervision, or an external ground-truth oracle (a verifier, a simulator, the physical world)? If it does, it can raise the ceiling. If it merely re-processes what its own models already contain — however cleverly — our results predict it will **plateau**. Recursive self-improvement compounds precisely where an external oracle exists to be exploited (a game's rules, a theorem checker, a compiler, a market, a lab), and stalls where one does not. That is a claim about the *source* of information, not the ingenuity of the method — and it should hold well beyond chess. - [chessnet/model.py](#) — conv/MLP/dual-path + value head. [chessnet/search.py](#) — alpha-beta, MCTS/PUCT, quiescence, the wide→narrow cascade ([MultiStageMCTSPlayer](#)). [chessnet/committee.py](#) — ensemble inference + agreement signal. [chessnet/train.py](#) — soft/hard + value training. [scripts/selfplay.py](#) — self-play iteration. - **Best model:** [runs/conv_value_llm1](#) (conv-96×8 + value, 3.45M params). - **Key hyperparameters:** conv width 96, depth 8; lr 5e-4 (train) / 1e-4 (self-play); gradient clip 1.0; MCTS c_puct 1.5; Dirichlet α 0.3; replay buffer 120K–300K.